

## Lab 1 — Code Hint Sheet

### Your First Network: the PyTorch training loop

Stuck on a **YOUR TURN** cell? Each line you need is below with a few options. Pick one, type it in, and **run the cell** — the built-in `assert` checks will tell you whether it is right. Only the two marked cells need editing; run everything else as-is.

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**Step 4 — the training loop.** Write these five lines *in order* inside the `for epoch` loop (Lecture 2: forward → loss → zero → backward → step).

**1. Forward — make a guess.**

- (a) `pred = model(X_train)`
- (b) `pred = loss_fn(X_train)`
- (c) `pred = model.guess(y_train)`

**2. Loss — how wrong?**

- (a) `loss = loss_fn(pred, y_train)`
- (b) `loss = model(pred, y_train)`
- (c) `loss = loss_fn(X_train, y_train)`

**3. Zero — clear last step's gradients.**

- (a) `optimizer.zero_grad()`
- (b) `optimizer.reset()`
- (c) `model.zero_grad(loss)`

**4. Backward — backprop the blame.**

- (a) `loss.backward()`
- (b) `optimizer.backward()`
- (c) `model.backward(loss)`

**5. Step — nudge every weight.**

- (a) `optimizer.step()`
- (b) `optimizer.update()`
- (c) `loss.step()`

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**Step 6 — evaluate on the test set.** Predict without learning, threshold at 0.5, then measure the fraction correct.

**1. Predict without gradients.**

- (a) `with torch.no_grad():` (then, indented) `probs = model(X_test)`
- (b) `with torch.grad():` (then, indented) `probs = model(X_test)`
- (c) `probs = model.no_grad(X_test)`

**2. Turn probabilities into 0/1 predictions.**

- (a) `predicted = (probs > 0.5).float()`
- (b) `predicted = probs.round(threshold=0.5)`
- (c) `predicted = probs.argmax()`

**3. Accuracy — fraction that match the label.**

- (a) `test_acc = (predicted == y_test).float().mean().item()`
- (b) `test_acc = (predicted - y_test).mean()`
- (c) `test_acc = accuracy(predicted, y_test)`

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Answers self-check in the notebook: if the `assert` cells print their success message, you chose right. The loss should fall steeply and then flatten, and test accuracy should land around **0.72–0.77** (coin-flip baseline is 0.50).